

Table 1: Summary of studies employing aerial remote sensing using visible band imagery only for large wood detection and quantification in rivers and river deltas. Many quantities have been estimated due to a lack of reporting of the dimensions of regions mapped, image pixel size, and other important details. See also Table 1 in Sendrowski and Wohl (2021), which lists other studies utilizing elevation-based and hyper-spectral imagery methods.

Study	Data source	Site	Mapped active channel (ha)	Number of surveys (total time)	Wood detection methods	Mapped units	Pixel horizontal footprint (m)
Marcus et al. (2002)	Fixed wing plane aerial nadir imagery (estimated)	Yellowstone, USA	~ 100 ha	Not clear. Several surveys on several rivers	Semi-automated (details lacking)	Large wood	Not reported, estimated < 1 m
Haschenburger and Rice (2004)	Fixed wing plane aerial nadir imagery	Carnation Creek, BC, Canada	~63 ha	One (details lacking)	Manual	Large Wood	Not reported, estimated < 1 m
Smikrud and Prakash (2006)	Fixed wing plane aerial nadir imagery	Unuk River, AK, USA	32 ha	Two (2003 – 2004)	Semi-automated supervised classification	Large Wood	0.8 m
Lassette et al. (2008)	Aerial imagery	Ain River, France	~800 ha	Several (1945 to 2000)	Manual	Large Wood	Not reported, estimated < 1 m
Ortega-Terol et al. (2014)	Aerial imagery	Jucar River, Spain	~ 2640 ha	One	Manual, semi-automated supervised classification	Large Wood	0.25 m
Atha (2014)	Satellite	Queets River, WA, USA	~ 730 ha	One (composite of public satellite imagery)	Manual	Large Wood	Not reported, estimated < 10 m
Ulloa et al. (2015)	Google Earth imagery, aerial imagery, and satellite image	Blanco River, Chile	~ 102 ha	Three (2005, 2009, 2012.)	Manual and semi-automated supervised classification	Large Wood	0.5 – 2.4 m

Taminga et al. (2015)	Orthoimagery from small UAV (quadcopter)	Elbow River, Alberta, Canada	30 ha	One (Sept. 2012)	Manual	Large Wood	0.05 m
Boivin et al. (2015)	Satellite and aerial imagery	St. Jean River, Quebec, Canada	~ 500 ha	Numerous (1963 – 2013)	Manual	Large Wood	0.63 – 15 m
Kramer et al. (2017)	Satellite and aerial imagery	Slave River, NWT, Canada	4 sample locations in an estimated 6489 ha region	59 (1983 – 2014)	Manual and semi-automated supervised classification	Large Wood	Not reported, estimated < 10 m
Sanhueza et al. (2019)	Orthoimagery from small UAV (quadcopter)	Blanco River, Chile	1.2 ha	One (Mar-Apr 2016)	Manual	Large Wood	0.016 m
Tsunetaka et al. (2021)	Orthoimagery from small UAV (quadcopter)	Otoishi River, Kyushu, Japan	35.2 ha	Two (2018 – 2019)	Manual	Large Wood	0.03 – 0.05 m
Sendrowski et al. (2023)	Satellites (GeoEye, Quickbird, and WorldView)	Mackenzie River Delta, AK, USA	35,000 ha	Not reported (combined 2009 – 2021)	Manual delineation to train fully automated deep learning model. Manual post-processing.	Large Wood	0.3 – 0.65 m
Iroumé et al. (2023)	Orthoimagery from small UAV (quadcopter)	Blanco Este River, Chile	10.9 ha	17 (2017-2020)	Manual	Large Wood	0.02 m
This study	Orthoimagery from fixed wing airplane	Elwha River, WA, USA	246 ha	14 (2012 – 2017)	Manual and semi-automated delineation to train fully automated deep learning model	Large Wood, Sediment, Water, Vegetation	0.125 m

	Overall accuracy	Frequency weighted Intersection over Union	Mean Intersection over Union	Matthews Correlation Coefficient
Wood (binary)	0.98	0.96	0.67	0.49
Sediment (binary)	0.95	0.92	0.79	0.73
Water (binary)	0.99	0.98	0.84	0.28
Other (binary)	0.99	0.97	0.88	0.65
Multiclass (wood, sediment, water, vegetation)	0.95	0.91	0.63	0.69

Table 2: Image semantic segmentation model accuracy statistics. Each row represents a different trained model, and each numeric column summarizes a different accuracy statistic.

	F1 score	Recall	Precision
Vegetation	0.98	0.99	0.99
Water	0.90	0.91	0.92
Sediment	0.72	0.74	0.75
Wood	0.60	0.81	0.53

Table 3: Multiclass segmentation model accuracy statistics. Each row represents a different class, and each numeric column summarizes a different accuracy statistic.

	Correlation coefficient
Wood abundance (time), MR and LR	0.78
Sediment abundance (time), MR and LR	0.89
Wood and sediment abundance (time), MR	0.86
Wood and sediment abundance (time), LR	0.77
Wood and sediment abundance (space), MR	0.88
Wood and sediment abundance (space), LR	0.92

Table 4: Correlations between distributions of wood and sediment in the middle reach (MR) and lower reach (LR).